

Educational Dataset Construction Pipeline

Detailed methodology for transforming educational textbooks into a structured dataset for Educational Knowledge Graph construction and knowledge-gap analysis.

Purpose: This document describes the current proposed pipeline. The pipeline constructs a reusable dataset before Knowledge Graph construction. The final dataset contains two files: **concepts.csv** and **relations.csv**.

1. Overall Architecture

Educational Textbook → Document Parsing → Educational Structure Detection → Structure-Aware Chunking → Concept Extraction → Concept Normalization → Definition Extraction → Relation Extraction → Relation Classification → Educational Weight Assignment → Dataset Construction → {concepts.csv, relations.csv} → Educational Knowledge Graph → Knowledge Gap Detection

2. Step-by-Step Methodology

Step 1: Document Parsing

Convert the textbook into machine-readable content while preserving useful structure.

- Text content
- Page numbers
- Chapter and section headings
- Paragraph boundaries
- Tables and formulas

Step 2: Educational Structure Detection

Identify meaningful educational blocks such as chapters, sections, definitions, explanations, examples, exercises, theorems, proofs, algorithms, and summaries. Explicit headings and layout cues can be detected using rules; ambiguous blocks may use semantic classification.

- Chapter
- Section
- Definition
- Explanation
- Example
- Exercise
- Theorem
- Proof
- Algorithm
- Summary

Step 3: Structure-Aware Educational Chunking

Divide the textbook according to educational boundaries instead of arbitrary fixed token windows. A definition should remain intact where possible, while examples and exercises can become separate chunks.

- Unique chunk ID
- Chapter and section context
- Page information
- Educational boundaries

Step 4: Concept Extraction

Identify important educational concepts from each chunk. A hybrid approach can generate candidates using NLP or keyword methods and selectively use an LLM for semantic refinement.

- Gradient Descent
- Loss Function
- Neural Network
- Hidden Layer

Step 5: Concept Normalization and Entity Linking

Merge equivalent expressions and assign one canonical concept name and a unique concept ID.

- Gradient Descent
- GD
- Gradient Descent Algorithm → Canonical concept: Gradient Descent

Step 6: Definition Extraction

Identify the best textbook-supported definition for each normalized concept. Definitions should be grounded in source content rather than freely generated.

- Concept: Gradient Descent
- Definition: An optimization algorithm used to minimize a loss function.

Step 7: Relation Extraction

Identify relationships between concepts from educational content. Preserve the source sentence as evidence.

- Example: Neural Networks are trained using Gradient Descent.
- Raw relation: Neural Network — trained using — Gradient Descent

Step 8: Relation Classification

Map raw natural-language relationships into a controlled ontology containing five predefined relation types.

- Prerequisite
- Part-of
- Application
- Extension
- Similarity

Step 9: Educational Weight Assignment

Assign each accepted relation a numerical weight representing its strength or educational importance. The exact weighting methodology should be explicitly defined and experimentally validated. This weight is not the same as LLM confidence.

- Weighted graph edge
- Educational relationship strength

Step 10: Dataset Construction

Store the processed concepts and relationships in two reusable CSV files. These files become the structured dataset used for downstream Knowledge Graph construction.

- concepts.csv
- relations.csv

3. Controlled Relation Ontology

Restricting relationships to a controlled ontology improves consistency and makes graph traversal easier.

Relation	Meaning and Direction
Prerequisite	$A \rightarrow B$ means A should be understood before B.
Part-of	$A \rightarrow B$ means A is a component or part of B.
Application	$A \rightarrow B$ means A is applied or used within B.
Extension	$A \rightarrow B$ means A builds upon or extends B.
Similarity	$A \leftrightarrow B$ means A and B are conceptually similar and may be treated as symmetric.

4. Final Dataset Structure

The current design intentionally keeps the dataset simple and contains only two CSV files.

A. concepts.csv

Column	Purpose
concept_id	Unique identifier for each canonical concept.
concept_name	Canonical name of the concept.
aliases	Alternative names or equivalent expressions.
concept_type	Type such as Algorithm, Model, Formula, Theory, or Component.
definition	Textbook-supported definition.
chapter	Chapter containing the source occurrence.
section	Section containing the source occurrence.
page	Source page number.
chunk_id	Identifier of the structure-aware source chunk.
evidence	Exact source text supporting the extracted concept.

B. relations.csv

Column	Purpose
relation_id	Unique identifier for each relation.
source_id	Concept ID from which the relation originates.
target_id	Concept ID to which the relation points.
relation_type	Prerequisite, Part-of, Application, Extension, or Similarity.
weight	Strength or educational importance of the relationship.
page	Page containing supporting evidence.
chunk_id	Identifier of the source chunk.
evidence	Exact source text supporting the relationship.

5. Example Relations

Source	Target	Relation Type	Meaning
Calculus	Gradient Descent	Prerequisite	Calculus supports understanding Gradient Descent.
Hidden Layer	Neural Network	Part-of	A Hidden Layer is a component of a Neural Network.
Gradient Descent	Neural Network	Application	Gradient Descent is used in training Neural Networks.
Advanced Gradient Method	Gradient Descent	Extension	The advanced method builds upon Gradient Descent.

Linear Regression	Logistic Regression	Similarity	The concepts share conceptual or structural similarities.
-------------------	---------------------	------------	---

6. Role of the LLM

The pipeline is not intended to depend completely on an LLM. Deterministic processing should be used wherever possible, while an LLM is selectively used for tasks requiring semantic reasoning.

Component	Preferred Approach
Document parsing	Parser and deterministic document processing
Structure detection	Rules and layout analysis; semantic fallback when ambiguous
Chunking	Structure-aware deterministic algorithm
Concept candidates	NLP/keyword extraction with optional LLM refinement
Concept normalization	Similarity matching and entity linking
Definition extraction	Source retrieval with grounded semantic processing
Relation extraction	Semantic extraction with selective LLM support
Relation classification	Controlled ontology mapping with semantic support
Dataset generation	Deterministic CSV construction

7. Evidence and Traceability

Every concept and relation should remain traceable to the original textbook. Page number, chunk ID, and evidence text are preserved so that extracted knowledge can be checked and explained later.

8. Downstream Use

- **concepts.csv** provides the nodes of the Educational Knowledge Graph.
- **relations.csv** provides the edges and their weights.

The resulting graph can support user-prompt concept mapping, localized graph exploration, concept coverage analysis, latent knowledge-gap identification, and explainable knowledge recommendations.

9. Proposed Research Contribution

Proposed contribution statement: The proposed system introduces a structure-aware educational dataset construction framework that transforms textbook content into a reusable concept-and-relation dataset. The framework preserves educational document structure, performs concept extraction and normalization, maps semantic relationships to a controlled educational ontology, and generates weighted relations with source evidence for downstream Educational Knowledge Graph construction and knowledge-gap analysis.

10. Current Scope

The current dataset deliberately focuses on the information essential for Knowledge Graph construction. Educational role and confidence/trust scores have been removed from the final CSV design for now. They can later be explored as extensions if experimental evaluation shows that they improve extraction quality or knowledge-gap ranking.